

Titan's Arena

Titan's Arena is a single-arena 3D game divided into three progressive levels, where each level introduces new gameplay rules without changing the environment. The player has three lives and must escape the arena by overcoming escalating threats. In Level 1, the player fights zombies as a warm-up phase and must kill at least five zombies, all of which die in one hit. In Level 2, inspired by Squid Game's Red Light–Green Light, the zombies disappear, and a giant Titan appears frozen at the center while the player's weapon is disabled; the player may move only during Green Light and must freeze during Red Light, as any movement during Red Light triggers a laser attack from the Titan. The objective is to reach a designated weapon zone to unlock the weapon. In Level 3, the Titan becomes fully active and attacks the player using a line-of-sight laser system. The arena contains four pillars that act as cover but are destructible after three laser hits. The Titan follows a 5-second attack phase and a 3-second cooldown phase during which its core is exposed, allowing the player to damage it. After five successful core hits, the Titan is defeated, and the player can escape the arena to win the game.

Features–

1. Single-Arena, Multi-Level Design

- One 3D arena is reused across all gameplay stages
- Level progression achieved by changing rules, not scenes

2. Three-Level Progression System

- Level 1: Zombie combat warm-up
- Level 2: Rule-based Red Light–Green Light challenge
- Level 3: Line-of-sight Titan boss fight

3. Rule-Based Red Light–Green Light Mechanic

- Timer-controlled switching between movement-allowed and movement-restricted states
- Player movement detection using position comparison
- Immediate punishment via laser for rule violation

4. Line-of-Sight Laser System

- Titan fires a laser using raycasting toward the player
- Laser interaction checks collisions with pillars and player
- Closest collision determines outcome

5. Finite State Machine (FSM) for Boss Behavior

- Titan transitions between Attack, Cooldown, and Defeated states
- State transitions controlled by timers and player actions

6. Destructible Environment Mechanics

- Four pillars act as dynamic cover
- Each pillar degrades after laser hits and eventually disappears
- Forces continuous player repositioning

7. Conditional Vulnerability (Boss Core System)

- Titan core exposed only during cooldown phase
- Player can damage boss only during this window
- Encourages timing and strategic play

8. Player Progression & Weapon Unlock System

- Weapon disabled during Level 2
- Weapon unlocked only after reaching the designated zone
- Creates meaningful progression between levels

9. Visual State Feedback System

- Color-coded indicators (Red/Green/Blue) for Titan states
- On-screen HUD text for rules, level, and warnings

10. Clear Win Condition & Ending Sequence

- Boss defeat animation
- Exit door activation
- Final success message displayed